6

The Response of a Medium

Preamble

There are two different descriptions of the electromagnetic response of a medium. One, which is the older description, involves introducing the polarization and magnetization of the medium and was originally applied to the response to static fields. This method may be generalized to apply to fluctuating fields but it becomes cumbersome and ill-defined for sufficiently general media. The other description is based on the use of Fourier transforms and so applies only to fluctuating fields. The Fourier transform description is used widely in plasma physics, and although it is less familiar than the other description when applied to dielectrics and magnetizable media, it is simpler and no less general than the older description.

6.1 Static Responses

The response of a medium to a static uniform electromagnetic field is described in terms of induced dipole moments. On a microscopic level a static uniform electric field polarizes the atoms or molecules. The *polarization* $\mathbf{P}$ is defined as the induced electric dipole moment per unit volume. A medium which becomes polarized in this way is called a *dielectric.*

The response of a magnetizable medium to a static uniform magnetic field is attributed to induced magnetic dipole moments and is described in terms of the *magnetization* $\mathbf{M}$, which is defined as the induced magnetic dipole moment per unit volume. Magnetizable media are classified

as paramagnetic or diamagnetic depending on whether the magnetization is parallel or antiparallel, respectively, to the applied magnetic field. ("Ferromagnetism" is regarded as an extreme form of paramagnetism in which the magnetization remains when the inducing magnetic field is turned off.) From a classical point of view one expects the induced current to obey Lenz' law and be such that the associated magnetic field opposes the original field; this would imply that all media should be diamagnetic, contrary to the existence of paramagnetism and ferromagnetism. Paramagnetism is associated with the alignment of the intrinsic magnetic dipoles (due to the spin of the electron) in the medium by the imposed magnetic field.

The description in terms of induced dipole moments is strictly valid only for the case of static uniform fields. The assumption that the fields are static implies a complete separation of electric and magnetic effects. Suppose that we relax the assumption that the imposed field is uniform, but still assume that it is static (the field then has a spatial derivative but not a temporal derivative). The responses can then include higher order multipole moments and magnetoelectric responses (a polarization induced by a magnetic field or a magnetization induced by an electric field). These more general responses, discussed in §6.2, are excluded for uniform fields simply because, for example, a quadrupole moment is induced by the gradient in the field, and an octupole moment is induced by the second derivative of the field. For non-uniform static fields the response in terms of $\mathbf{P}$ and $\mathbf{M}$ remains approximately valid for any medium in which the multipole expansion converges sufficiently rapidly, as is the case for many media.

In a similar way, the static response remains approximately valid when the assumption that the fields are static is relaxed to allow sufficiently slowly varying fields. However, this raises the question as to when a field is to be regarded as static or slowly varying, and when it is to be regarded as fluctuating. In practice, the distinction between slowly varying and fluctuating fields involves a separation of time scales, and no general condition can be specified. Usually there are natural frequencies associated with a medium, and a field is regarded as slowly varying if it varies on a time scale that is long compared with the inverse of the lowest relevant natural frequency of the medium. An extreme example is that of a vacuum in quantum electrodynamics where the response is due to virtual electron–positron pairs; the only natural frequency is then that obtained by dividing the rest energy mc^2 of the electron by Planck's constant $\hbar$, and this corresponds to a γ-ray energy of 0.511 MeV, so

that even fields at optical frequencies (energies $\lesssim 1\,\mathrm{eV}$) and higher are regarded as slowly varying in this case. Another extreme example is an electrolyte (a solution containing free ions) in which case there is no static electric response (in the sense implied by the foregoing discussion) because all the ions are freely accelerated by a static electric field; this acceleration is limited by collisions so that all ions reach a terminal drift velocity and the response in this case is a constant current and not a polarization.

For many media there is at least an approximate proportionality between the response and the imposed field. When such a proportionality is assumed to apply, the medium is said to be *linear*. The linear response of a medium is described by a response function defined as the constant of proportionality between the response and the imposed field.

It is conventional to introduce two additional fields when describing a static response. These are the *electric induction* (or *electric displacement*) $\mathbf{D}$ and the *magnetic field strength* $\mathbf{H}$. They are defined by writing

$$\mathbf{D} := \varepsilon_0\mathbf{E} + \mathbf{P}, \quad \mathbf{H} := \mathbf{B}/\mu_0 - \mathbf{M}. \tag{6.1}$$

The traditional description of the response of an isotropic medium to static fields is through the relations

$$\mathbf{D} = \varepsilon\mathbf{E}, \quad \mathbf{H} = \mathbf{B}/\mu, \tag{6.2}$$

where the response functions are the *dielectric permittivity* ε and the *magnetic permeability* μ . Alternatively, the response is described by

$$\mathbf{P} = \varepsilon_0\chi^{\mathrm{e}}\mathbf{E}, \quad \mathbf{M} = \chi^{\mathrm{m}}\mathbf{B}/\mu_0, \tag{6.3}$$

where χ^{e} is the *electric susceptibility* and χ^{m} is related to the *magnetic susceptibility*. (The magnetic susceptibility is defined as $\mu/\mu_0 - 1$, due to the disturbance being described by the field $\mathbf{H}$ rather than $\mathbf{B}$ in (6.3); this is an unfortunate legacy of an outdated convention.) The *dielectric constant* K, which is dimensionless, is defined by writing

$$\varepsilon = K\varepsilon_0, \quad K = 1 + \chi^{\mathrm{e}}. \tag{6.4}$$

A corresponding dimensionless constant is sometimes defined for the magnetic response.

The underlying procedure for describing the response in terms of $\mathbf{P}$ and $\mathbf{M}$ involves a multipole expansion truncated at the dipole terms. Quite generally, the response may be described in terms of the induced charge density ρ_{ind} and the induced current density $\mathbf{J}_{\mathrm{ind}}$. These are related to the polarization and magnetization by

$$\rho_{\mathrm{ind}} = -\mathrm{div}\,\mathbf{P}, \quad \mathbf{J}_{\mathrm{ind}} = \partial\mathbf{P}/\partial t + \mathrm{curl}\,\mathbf{M}. \tag{6.5}$$

In a theory where $\mathbf{P}$ and $\mathbf{M}$ are defined, (6.5) is a prescription for constructing ρ_{ind} and $\mathbf{J}_{\text{ind}}$. In other theories where ρ_{ind} and $\mathbf{J}_{\text{ind}}$ are calculated in an independent way, (6.5) defines $\mathbf{P}$ and $\mathbf{M}$, but the definition is incomplete and ambiguous.

To see how (6.5) arises, note first that the electric dipole moment $\mathbf{d}(t)$ and the magnetic dipole moment $\mathbf{m}(t)$ of any charge and current distribution are defined by (3.8) and (3.12), *viz.*,

$$\mathbf{d}(t) = \int \mathrm{d}^3\mathbf{x}\,\mathbf{x}\,\rho(t,\mathbf{x}), \quad \mathbf{m}(t) = \tfrac{1}{2}\int \mathrm{d}^3\mathbf{x}\,\mathbf{x}\times\mathbf{J}(t,\mathbf{x}), \tag{6.6}$$

respectively. These are leading terms in a multipole expansion which is related to the spatial Fourier transform by expanding the exponentials in

$$\rho(t,\mathbf{k}) = \int \mathrm{d}^3\mathbf{x}\,\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\,\rho(t,\mathbf{x}), \tag{6.7}$$

$$\mathbf{J}(t,\mathbf{k}) = \int \mathrm{d}^3\mathbf{x}\,\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\,\mathbf{J}(t,\mathbf{x}). \tag{6.8}$$

That is, one makes the expansion

$$\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}} = 1 - \mathrm{i}\mathbf{k}\cdot\mathbf{x} + \tfrac{1}{2}(\mathrm{i}\mathbf{k}\cdot\mathbf{x})^2 + \cdots. \tag{6.9}$$

On inserting the expansion (6.9) into (6.7), the unit term in (6.9) gives the net electric charge, which is assumed to be zero here, the linear term is related to the electric dipole moment, the quadratic term to the electric quadrupole moment, and so on. On inserting the expansion (6.9) into (6.8), the unit term is related to the time-derivative of the electric dipole moment, cf. (3.14), the symmetric and antisymmetric parts of the next term are related to the electric quadrupole moment and magnetic dipole moment, respectively, cf. (3.15), and so on. Then on writing $\mathbf{d}(t) = \int \mathrm{d}^3\mathbf{x}\,\mathbf{P}(t,\mathbf{x})$ and $\mathbf{m}(t) = \int \mathrm{d}^3\mathbf{x}\,\mathbf{M}(t,\mathbf{x})$, one finds that if only the electric and magnetic dipole terms are retained, then (6.7) and (6.8) with (6.9) are approximated by

$$\rho(t,\mathbf{k}) = \int \mathrm{d}^3\mathbf{x}\,[-\mathrm{i}\mathbf{k}\cdot\mathbf{P}(t,\mathbf{x})], \tag{6.10}$$

$$\mathbf{J}(t,\mathbf{k}) = \int \mathrm{d}^3\mathbf{x}\,[\partial\mathbf{P}/\partial t + \mathrm{i}\mathbf{k}\times\mathbf{M}]. \tag{6.11}$$

In evaluating the electric dipole term in (6.11) one needs to use the charge continuity equation and to partially integrate (ignoring surface terms). Finally on inverting the spatial Fourier transform, (6.10) and (6.11) lead to (6.5).

The foregoing derivation involves approximations that are not valid in general. It is not obvious that higher order terms in the expansion (6.9)

can be neglected, and indeed in some media it is important to retain the electric quadrupole term, as discussed in §6.2. Moreover, there is no guarantee that the multipole expansion converges rapidly, or that it converges at all. There is also a more subtle difficulty: the multipole expansion is incomplete because certain terms in the expansion (6.9) are omitted; specifically the traces of all multipole moments are defined to be zero, cf. §3.2. The traces are not zero in general, and even if the multipole expansion does converge, it needs to be supplemented by the trace terms for the result to be completely general.

One case where the multipole expansion does not converge is in an electrolyte or in a thermal plasma. Then the Debye shielding effect causes the response of the medium to become arbitrarily large at arbitrarily small distances, that is, at arbitrarily large k. The multipole expansion is based on (6.9), and applies only for $|\mathbf{k}\cdot\mathbf{x}| \ll 1$, which corresponds to sufficiently small k. This expansion is invalid in the large-k limit required to treat Debye shielding.

6.2 Properties of Media

The electromagnetic properties of media are reflected in mathematical properties of the response functions. The simplest media, in the sense of §6.1, are described by their dielectric permittivity ε or dielectric constant K and by their magnetic permeability μ. Actual media are often more complicated and may exhibit any of the following phenomena: non-linearity, anisotropy, dispersion, optical activity, spatial dispersion, gyrotropy, and magnetoelectric responses. Let us consider each of these briefly.

Non-linearity: Non-linear effects, such as hysteresis, may be included in the response functions for sufficiently simple media. However, in general one needs to expand in the electromagnetic field and to treat the linear and non-linear terms separately. An expansion of the response in powers of the applied field defines a hierarchy of non-linear response functions, cf. §6.4. In the following summary it is assumed that only the linear response is retained.

Anisotropy: A medium is anisotropic if the response is not in the same direction as the disturbance. Then relations such as (6.3) need to be replaced by tensor relations:

$$P_i = \varepsilon_0 \chi^{\mathrm{e}}_{ij} E_j, \quad M_i = \chi^{\mathrm{m}}_{ij} B_j/\mu_0. \tag{6.12}$$

Thus the susceptibilities and other response functions become tensors.

Dispersion: Dispersion is the property that the response is a function of frequency. Dispersion is attributed to the existence of one or more natural frequencies in the response of the medium. Formally, dispersion is included by allowing all the quantities in (6.12) to be functions of ω. All media are necessarily dispersive, but for many media, such as glass, water and air, there is a broad range of natural frequencies, causing the dispersion to be weak, at least over the optical spectral range. Weak dispersion requires rather special conditions and in this sense it is surprising that so many familiar media are weakly dispersive. This is attributed to a selection effect: weak dispersion is related to transparency (dispersion and absorption are related through the Kramers–Kronig relations) and transparent materials are of obvious special interest in optics. Thus, dispersion is weak in transparent materials such as glass essentially because one selects materials that are transparent over a substantial frequency range, and this property is physically related to weak dispersion.

Dispersion is associated with the response at time t being due to disturbances at times $t' \neq t$. The *causal condition* is that cause must precede effect, and this requires that the response at t be due to disturbances only at $t' \leq t$. In the limit of arbitrarily slowly varying fields, the response becomes nearly instantaneous with the disturbance, and dispersion is then unimportant.

Natural Optical Activity: In some media the electric quadrupole response is significant, and this is the cause of a phenomenon called *natural optical activity.* Optically active media have an intrinsic handedness that is determined by the sense of rotation of the plane of polarization of light passing through the medium. The media are classified as *dextrorotatory* and *levorotatory.* The most familiar example of such a medium is a solution of dextrose. The inclusion of an electric quadrupole term implies a term in the response function that is linear in $\mathbf{k}$.

Spatial Dispersion: Spatial dispersion is the property that the response is a function of $\mathbf{k}$. Two specific types of spatial dispersion mentioned above are Debye shielding and natural optical activity; these involve specific functional forms of the dependence of the response functions on $\mathbf{k}$. More generally, the response function is regarded as an arbitrary function of $\mathbf{k}$ (as well as of ω if the medium is also dispersive). Spatial dispersion is associated with the presence of natural lengths in the medium, such as the dimensions of atoms, interatomic spacings or of inhomogeneities. Although all media are technically spatially dispersive, the effect is weak for wavelengths that are long compared with the natural lengths, that

is, for $kL \ll 1$, where $2\pi/k$ is the wavelength and L is a natural length in the medium.

Spatial dispersion is associated with non-local responses. That is, the response at a point $\mathbf{x}$ is due to disturbances at other points $\mathbf{x}' \neq \mathbf{x}$. The neglect of spatial dispersion implies a local response, such that the spatial variations of the response and the disturbance are identical.

An important point to note is that when discussing the response of a medium there is no implied relation between ω and $\mathbf{k}$. For electromagnetic waves and for other waves there is a relation between ω and $\mathbf{k}$ (the "dispersion relation" for the waves) but this does not imply any connection between dispersion and spatial dispersion. In principle, one studies the temporal response of a medium, and hence the dispersion, by subjecting it to a spatially uniform field that oscillates in time, and one studies the spatial response, and hence its spatial dispersion, by subjecting it to a static field that is spatially periodic. Dispersion and spatial dispersion are independent effects.

Gyrotropy: The presence of a magnetic field can cause a medium to exhibit the Faraday effect, that is, a rotation of the plane of polarization of radiation passing through the medium. Such media are said to be gyrotropic. Gyrotropy is an important feature of an ionized gas in a magnetic field, and is of particular relevance to the ionosphere, to astrophysical plasmas and to magnetically confined laboratory plasmas.

Magnetoelectric responses: In (6.3) the possibility is neglected that a magnetic disturbance may cause the medium to become polarized or an electric disturbance cause the medium to become magnetized. Such responses can occur in principle and media that exhibit them are said to be magnetoelectric. To include magnetoelectric responses two further terms need to be included in (6.12), a term relating the polarization to the magnetic field and a term relating the magnetization to the electric field.

A subtle point is that magnetoelectric responses correspond to a particular form of spatial dispersion. Once one has Fourier transformed in space (as well as in time) the magnetic field is completely specified by the transverse part of the electric field, and hence it is not possible to separate electric and magnetic disturbances in a unique way.

Other responses: There are further types of response that are not of direct relevance in the following discussion, but which should be mentioned. First there are modifications to high-frequency responses due to various kinds of stress imposed on a system. One example is the effect of a static electric field on the high-frequency response. For an

otherwise isotropic medium the effect is quadratic in the field, and leads to a response tensor of the form $K_{ij}(\omega) = K^{(0)}(\omega)\delta_{ij} + \beta_{ij}^{(\mathbf{E})}(\omega)E_iE_j$, implying that the medium becomes uniaxial with principal axis along $\mathbf{E}$. This is called the *Kerr effect.* The corresponding magnetic effect, in which the static $\mathbf{E}$ is replaced by a static $\mathbf{B}$ is the *Cotton–Mouton effect.* There are analogous modifications to $K_{ij}(\omega)$ due to elastic deformation of the medium, to sheared velocity flows in a fluid medium (*streaming birefringence* or the *Maxwell effect*), and so on. One such effect that is important in the following discussion is a modification to $K_{ij}(\omega)$ that is linear in the static $\mathbf{B}$; this leads to the *Faraday effect*, which is attributed to the natural wave modes of the medium being elliptically (or circularly) polarized.

There are also other types of response in which only one of the disturbance and the response is electromagnetic in character. Perhaps the most familiar are the *thermoelectric effect*, in which an electric current flows when a system is subjected to a temperature gradient, and the *piezoelectric effect*, in which a crystal subjected to a mechanical stress becomes polarized. The existence of the inverse effects is implied by the Onsager relations. Thus a thermoelectric medium develops a heat flow in an electric field, and a piezoelectric medium develops a strain in the presence of an electric field.

6.3 Two Alternative Descriptions of the Linear Response

There are two different ways of introducing a linear response function for an arbitrary medium. One is based on generalizing the response functions in (6.3), as outlined above. The other involves starting from the Fourier transform of the induced current and introducing a single response tensor that is a function of ω and $\mathbf{k}$.

The most general form of a linear response in terms of the polarization and the magnetization is for a medium which is anisotropic, dispersive, spatially dispersive and magnetoelectric. This general form involves the relations

$$\begin{aligned} P_i(\omega,\mathbf{k}) &= \chi_{ij}^{\mathrm{e}}(\omega,\mathbf{k})\varepsilon_0 E_j(\omega,\mathbf{k}) + \chi_{ij}^{\mathrm{em}}(\omega,\mathbf{k})B_j(\omega,\mathbf{k})/\mu_0 c, \\ M_i(\omega,\mathbf{k}) &= \chi_{ij}^{\mathrm{me}}(\omega,\mathbf{k})\varepsilon_0 c E_j(\omega,\mathbf{k}) + \chi_{ij}^{\mathrm{m}}(\omega,\mathbf{k})B_j(\omega,\mathbf{k})/\mu_0. \end{aligned} \tag{6.13}$$

Thus when anisotropy, dispersion and spatial dispersion, and magnetoelectric responses are included one appears to require four second-rank tensors that are functions of ω and $\mathbf{k}$ to describe a general response using this theory. The Onsager relations, cf. §7.4, imply a relation between the magnetoelectric response tensors χ_{ij}^{em} and χ_{ij}^{me}, so that only three different 3-tensors need be introduced.

Although (6.13) is a general form of the response of a medium to fluctuating fields, it is ill defined. As mentioned above, magnetoelectric responses may be regarded as a specific form of spatial dispersion, and the inclusion of magnetoelectric response tensors that are functions of $\mathbf{k}$ as well as of ω introduces redundancy. Strictly, a magnetoelectric medium is one in which the response is of the form (6.13) with all the response tensors independent of $\mathbf{k}$.

One difficulty with (6.13) is that the electric field $\mathbf{E}(\omega, \mathbf{k})$ already includes the magnetic field $\mathbf{B}(\omega, \mathbf{k})$ through (5.5), and hence it is not clear how the separation into electric and magnetic fields is to be made on the right hand side of (6.13). Thus, once one has Fourier transformed in both space and time, the separation into electric and magnetic disturbances is ill defined. Also, the separation of the induced current into electric and magnetic parts in the Fourier transform of (6.5) is ill defined, that is,

$$\mathbf{J}_{\text{ind}}(\omega, \mathbf{k}) = -\mathrm{i}\omega\mathbf{P}(\omega, \mathbf{k}) + \mathrm{i}\mathbf{k} \times \mathbf{M}(\omega, \mathbf{k}) \tag{6.14}$$

does not specify how the separation into $\mathbf{P}(\omega, \mathbf{k})$ and $\mathbf{M}(\omega, \mathbf{k})$ is to be made. In particular one could define $\mathbf{M}(\omega, \mathbf{k})$ to be zero without loss of generality; then all the response would be included in $\mathbf{P}(\omega, \mathbf{k})$. As the full disturbance is already included in $\mathbf{E}(\omega, \mathbf{k})$, (6.13) is replaced by a single equation which relates $\mathbf{P}(\omega, \mathbf{k})$, now defined by (6.14) with $\mathbf{M}(\omega, \mathbf{k})$ identically zero, to $\mathbf{E}(\omega, \mathbf{k})$. This form is equivalent to the description of the response in the alternative method discussed below.

The use of the polarization and magnetization to describe the response of a medium is useful only for media which are not spatially dispersive or which have particular forms of spatial dispersion, such as in natural optical activity.

The basis for this alternative description is that, after Fourier transforming, both the disturbance and the response are each described completely in terms of a single electric field. (Note that this excludes static fields because they are excluded on Fourier transforming.) A general response is described completely in terms of the relation between the Fourier transform of the induced current density $\mathbf{J}_{\text{ind}}(\omega, \mathbf{k})$ and the electromagnetic disturbance as described by $\mathbf{E}(\omega, \mathbf{k})$. The choice of $\mathbf{E}(\omega, \mathbf{k})$ leads to a relation

$$(J_{\text{ind}})_i(\omega, \mathbf{k}) = \sigma_{ij}(\omega, \mathbf{k}) E_j(\omega, \mathbf{k}), \tag{6.15}$$

where $\sigma_{ij}(\omega, \mathbf{k})$ is the *conductivity tensor*. The point that is to be emphasized is that (6.15) contains the same information as (6.13). The seemingly greater generality of (6.13) is an illusion because neither the

separation of the disturbance into $\mathbf{E}(\omega, \mathbf{k})$ and $\mathbf{B}(\omega, \mathbf{k})$ nor the separation of the disturbance into $\mathbf{P}(\omega, \mathbf{k})$ and $\mathbf{M}(\omega, \mathbf{k})$ is properly defined when one Fourier transforms in space as well as in time.

The conductivity tensor in the general sense used in (6.15) includes both non-dissipative and dissipative processes. The more familiar use of "electrical conductivity" or its inverse the "resistivity" to describe ohmic dissipation of electric currents refers only to the dissipative part of the conductivity in the general sense used in (6.15). In fact the "electrical conductivity" corresponds to the low-frequency limit of the dissipative part of the electrical conductivity, cf. Exercise 6.3.

The choice of the vector potential $\mathbf{A}(\omega, \mathbf{k})$ in the temporal gauge in place of the electric field $\mathbf{E}(\omega, \mathbf{k})$ in (6.15) leads to a response of the form

$$(J_{\text{ind}})_i(\omega, \mathbf{k}) = \alpha_{ij}(\omega, \mathbf{k}) A_j(\omega, \mathbf{k}). \tag{6.16}$$

The tensor α_{ij} has no widely accepted name; *polarization response tensor* would reflect an analogy with the "vacuum polarization tensor" in quantum electrodynamics, however, in optics it is conventional to use "polarization tensor" with a different meaning in reference to the polarization of transverse waves. The relation (5.9), *viz.*, $\mathbf{E}(\omega, \mathbf{k}) = \mathrm{i}\omega\mathbf{A}(\omega, \mathbf{k})$, implies that $\alpha_{ij}(\omega, \mathbf{k})$ is equal to $\mathrm{i}\omega$ times $\sigma_{ij}(\omega, \mathbf{k})$.

In place of the conductivity tensor σ_{ij} or the polarization tensor α_{ij} it is often more convenient to use the *equivalent dielectric tensor* K_{ij} to describe the response. The equivalent dielectric tensor is defined to be dimensionless and to reduce to the unit tensor *in vacuo*. One defines K_{ij} by writing

$$\begin{aligned} K_{ij}(\omega, \mathbf{k}) &= \delta_{ij} + \frac{\mathrm{i}}{\omega\varepsilon_0}\sigma_{ij}(\omega, \mathbf{k}) \\ &= \delta_{ij} + \frac{1}{\omega^2\varepsilon_0}\alpha_{ij}(\omega, \mathbf{k}) \\ &= \delta_{ij} + \chi_{ij}(\omega, \mathbf{k}). \end{aligned} \tag{6.17}$$

The *equivalent permittivity tensor* $\varepsilon_{ij} = \varepsilon_0 K_{ij}$ differs from the dielectric tensor only in its dimensions. The forms (6.17) correspond to describing the response in terms of an equivalent polarization, which is defined by setting $\mathbf{M} = 0$ in (6.14), and then defining an equivalent electric induction through (6.1). The final form in (6.17) corresponds to introducing the equivalent susceptibility tensor $\chi_{ij}(\omega, \mathbf{k})$.

It should be emphasized that the equivalent dielectric tensor is not the same as the dielectric tensor introduced in (6.4). To emphasize the distinction between the description of the response in terms of Fourier transforms in space as well as time and the older method discussed above,

it is instructive to derive the equivalent dielectric tensor corresponding to the simple models of media described by (6.2), (6.3), (6.12) and so on. For example, the equivalent dielectric tensor that corresponds to (6.2) is

$$K_{ij}(\omega, \mathbf{k}) = (\varepsilon/\varepsilon_0)\,\delta_{ij} + (k^2c^2/\omega^2)\,(1 - \mu_0/\mu)\,(\delta_{ij} - \kappa_i\kappa_j), \tag{6.18}$$

with

$$\boldsymbol{\kappa} := \mathbf{k}/k, \quad k := |\mathbf{k}|. \tag{6.19}$$

A more general relation corresponds to the response described by (6.13) with all the susceptibilities functions of ω but not of $\mathbf{k}$; the corresponding equivalent dielectric tensor is

$$K_{ij}(\omega, \mathbf{k}) = \delta_{ij} + \frac{1}{\varepsilon_0}\left\{\chi^{\mathrm{e}}_{ij}(\omega) - \frac{k_r k_b c^2}{\omega^2}\,\epsilon_{irs}\chi^{\mathrm{m}}_{sa}(\omega)\epsilon_{abj} + \frac{k_s c}{\omega}\left[\chi^{\mathrm{em}}_{ir}(\omega)\epsilon_{rsj} - \epsilon_{isr}\chi^{\mathrm{me}}_{rj}(\omega)\right]\right\}. \tag{6.20}$$

It follows from (6.18) or (6.20) that the equivalent dielectric tensor is equal to the dielectric tensor, defined in the older sense, only for a dielectric which is not spatially dispersive and which has no magnetic properties. The equivalent dielectric tensor includes the magnetic as well as the electric response in general.

When the response is written in terms of Fourier transforms in space as well as in time, Maxwell's equations are reduced to a single equation called the *wave equation.* The wave equation is written down by making the separation into induced (ind) and extraneous (ext) parts of the current, $\mathbf{J} = \mathbf{J}_{\mathrm{ind}} + \mathbf{J}_{\mathrm{ext}}$ and including the induced part, as written in the form (6.16), on the left hand side of (5.8). The result is written in the form

$$\Lambda_{ij}(\omega, \mathbf{k})A_j(\omega, \mathbf{k}) = -\frac{\mu_0 c^2}{\omega^2}\,(J_{\mathrm{ext}})_i(\omega, \mathbf{k}), \tag{6.21}$$

with

$$\Lambda_{ij}(\omega, \mathbf{k}) := \frac{k^2c^2}{\omega^2}\,(\kappa_i\kappa_j - \delta_{ij}) + K_{ij}(\omega, \mathbf{k}). \tag{6.22}$$

6.4 Non-linear Response Tensors

Non-linear effects that involve waves are of wide interest in a variety of contexts. Examples include self-focusing of light, degenerate four-wave mixing and frequency doubling in optics and laser physics; cross modulation of radio waves in the ionosphere; wave decay, modulational instabilities and wave collapse in plasma physics; and photon splitting and

photon–photon scattering in quantum electrodynamics. These various effects are attributed to non-linear responses of the media in which they occur. Formally, non-linear response tensors are defined by expanding the response in powers of the disturbance. The form of this expansion is written down here and some qualitative remarks on its significance are made.

There are two natural starting points for a non-linear expansion. One is relevant to non-linear optics where spatial dispersion is unimportant. On expanding the polarization in powers of the electric field one obtains

$$P_i(\omega)/\varepsilon_0 = \chi_{ij}(\omega)\, E_j(\omega) + \int \mathrm{d}\omega^{(2)}\, \chi^{(2)}_{ijl}(\omega,\omega_1,\omega_2)\, E_j(\omega_1)\, E_l(\omega_2)$$
$$+ \int \mathrm{d}\omega^{(3)}\, \chi^{(3)}_{ijlm}(\omega,\omega_1,\omega_2,\omega_3)\, E_j(\omega_1)\, E_l(\omega_2)\, E_m(\omega_3) + \cdots, \tag{6.23}$$

with

$$\mathrm{d}\omega^{(n)} = \mathrm{d}\omega_1 \ldots \mathrm{d}\omega_n\, \delta(\omega - \omega_1 - \ldots - \omega_n), \tag{6.24}$$

and so on. The third-rank tensor $\chi^{(2)}_{ijl}(\omega,\omega_1,\omega_2)$ is the quadratic non-linear susceptibility and the fourth rank tensor $\chi^{(3)}_{ijlm}(\omega,\omega_1,\omega_2,\omega_3)$ is the cubic non-linear susceptibility. The δ-functions in (6.24) imply that $\chi^{(2)}_{ijl}$ is not an independent function of all three arguments, and that it is a function of only two frequency differences, for example, $\omega - \omega_1$ and $\omega - \omega_2$. Similarly, $\chi^{(3)}_{ijlm}$ is a function of three frequency differences.

The quadratic response allows the beating of two waves with frequencies ω_1 and ω_2 to form a third wave with frequency $\omega = \omega_1 + \omega_2$. The particular case of frequency doubling corresponds to $\omega_1 = \omega_2$. There is a kinematic constraint on such processes that can be very restrictive in practice: all waves must satisfy a beat condition $\mathbf{k} = \mathbf{k}_1 + \mathbf{k}_2$ as well as $\omega = \omega_1 + \omega_2$ and must also satisfy appropriate dispersion relations. Furthermore, the quadratic non-linear susceptibility is zero for many media. A subtle but powerful argument implies that $\chi^{(2)}_{ijl}$ is zero for an isotropic medium. The argument concerns the tensor indices. An arbitrary tensor may be written as a sum of terms consisting of scalar functions times tensorial factors constructed from the available vectors and tensors. For an isotropic medium there is no natural vector in the problem, except $\mathbf{k}$ and this is not available if the medium is not spatially dispersive (so that the response tensor does not depend on $\mathbf{k}$). Thus the tensor indices can appear only in the form of constant tensors that are always available; these are the unit tensor δ_{ij} or the permutation symbol ϵ_{ijl}. There is no way that one can construct a tensor with three indices from δ_{ij} with-

out involving a vector, and as there is no vector in the problem there can be no term constructed from δ_{ij}. It is apparent from (6.23) that $\chi^{(2)}_{ijl}(\omega, \omega_1, \omega_2)$ is unchanged by permuting the labels j, 1 and l, 2. This is inconsistent with $\chi^{(2)}_{ijl}$ being antisymmetric in j and l, and hence excludes any term that is dependent on ϵ_{ijl}. Thus there is no possible way of constructing an acceptable third-rank tensor for the quadratic non-linear response of an isotropic medium that is not spatially dispersive. In an anisotropic medium there is at least one characteristic direction associated with the medium. Examples include the principal axis of a uniaxial crystal and the normal to the plane of asymmetry in a medium which is not reflection symmetric. In such media $\chi^{(2)}_{ijl}$ is non-zero in general.

The term in (6.23) involving the cubic response tensor allows the existence of such processes as frequency trebling, degenerate four-wave mixing, wave–wave scattering and, depending on its sign, self-focusing of light. The cubic response tensor is non-zero for all media; it is even non-zero for the vacuum itself.

In a spatially dispersive medium, and in an ionized gas in particular, the so-called weak-turbulence expansion is of the induced current in powers of the amplitude of the Fourier transformed electromagnetic field. Choosing to describe the electromagnetic field in terms of the vector potential in the temporal gauge, this expansion has the form

$$\begin{aligned}(J_{\text{ind}})_i(k) &= \alpha_{ij}(k)\, A_j(k) + \int \mathrm{d}\lambda^{(2)}\, \alpha^{(2)}_{ijl}(k, k_1, k_2)\, A_j(k_1)\, A_l(k_2) \\ &\quad + \int \mathrm{d}\lambda^{(3)}\, \alpha^{(3)}_{ijlm}(k, k_1, k_2, k_3)\, A_j(k_1)\, A_l(k_2)\, A_m(k_3) + \cdots,\end{aligned} \tag{6.25}$$

where the nth order convolution integral is defined by

$$\mathrm{d}\lambda^{(n)} = \frac{\mathrm{d}^4k_1}{(2\pi)^4}\,\frac{\mathrm{d}^4k_2}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_n}{(2\pi)^4}\,(2\pi)^4\delta^4(k - k_1 - k_2 - \cdots - k_n), \tag{6.26}$$

with

$$\mathrm{d}^4k = \mathrm{d}\omega\mathrm{d}^3\mathbf{k}, \quad \delta^4(k) = \delta(\omega)\delta^3(\mathbf{k}). \tag{6.27}$$

The expansion (6.25) includes the linear response tensor $\alpha_{ij}(k)$, the quadratic response tensor $\alpha^{(2)}_{ijl}(k, k_1, k_2)$ and the cubic response tensor $\alpha^{(3)}_{ijlm}(k, k_1, k_2, k_3)$. Unlike the case of non-linear optics, the quadratic response tensor is non-zero for all plasmas, including isotropic plasmas. The important distinction between the non-linear response of a plasma and the non-linear response of media of interest in non-linear optics is that plasmas are spatially dispersive.

Exercise Set 6

6.1 A medium is described by the response functions $\varepsilon(\omega)$ and $\mu^{-1}(\omega)$ in

$$\mathbf{D} = \varepsilon(\omega)\mathbf{E}, \quad \mathbf{H} = \mu^{-1}(\omega)\mathbf{B}.$$

Construct the equivalent dielectric tensor $K_{ij}(\omega, \mathbf{k})$ in terms of $\varepsilon(\omega)$ and $\mu^{-1}(\omega)$. You should obtain the result (6.18).

6.2 Consider a medium in which the response is purely electrostatic and is described by the induced charge density $\rho(\omega, \mathbf{k})$. Suppose that a response function $X(\omega, \mathbf{k})$ is defined by writing

$$\rho(\omega, \mathbf{k}) = X(\omega, \mathbf{k})\phi(\omega, \mathbf{k}), \qquad (E6.1)$$

where $\phi(\omega, \mathbf{k})$ is the scalar potential in the Coulomb gauge. Find an expression that relates $X(\omega, \mathbf{k})$ to the components of the equivalent dielectric tensor $K_{ij}(\omega, \mathbf{k})$.

6.3 Let an electron gas be described as a fluid with a fluid velocity $\mathbf{v}$ and an equation of motion

$$m_{\mathrm{e}} \mathrm{d}\mathbf{v}/\mathrm{d}t = -e\mathbf{E} - \nu_{\mathrm{e}} m_{\mathrm{e}} \mathbf{v}, \qquad (E6.2)$$

where ν_{e} is a collision frequency.

(a) Construct the equivalent dielectric tensor $K_{ij}(\omega)$ for a cold unmagnetized plasma including the effects of collisions as follows. Fourier transform (E6.2), to find $\mathbf{v}(\omega)$, insert the result into

$$\mathbf{J}_{\mathrm{ind}}(\omega) = -en_{\mathrm{e}}\mathbf{v}(\omega),$$

and hence identify $\sigma_{ij}(\omega)$ and ω_{p} in

$$K_{ij}(\omega) = \delta_{ij}\left(1 - \omega_{\mathrm{p}}^2/\omega(\omega + \mathrm{i}\nu_{\mathrm{e}})\right). \qquad (E6.3)$$

(b) Show that in the limit $\omega \to 0$, $\sigma_{ij}(\omega)$ remains real and finite. Hence show that the electrical resistivity η, which is the inverse of the conductivity, is given by

$$\eta = \nu_{\mathrm{e}}/\varepsilon_0\omega_{\mathrm{p}}^2. \qquad (E6.4)$$

6.4 Consider an isotropic magnetoelectric medium whose response is described in the form (6.13) with all the susceptibilities of the form $\chi_{ij}(\omega, \mathbf{k}) = \chi(\omega)\delta_{ij}$.

(a) Use (2.24) in (6.22) to derive an explicit expression for the corresponding equivalent dielectric tensor.

(b) Show that if the medium has no magnetic properties ($\chi^{\mathrm{m}} = 0$) then its response is of the form

$$K_{ij}(\omega, \mathbf{k}) = K(\omega)\delta_{ij} + \eta(\omega)\epsilon_{ijr}k_r \qquad (E6.5)$$

that is characteristic of an optically active medium, and identify the rotatory part of the response $\eta(\omega)$.

(c) Does this imply that naturally optical active media are necessarily magnetoelectric and vice versa?

6.5 Derive the form (6.20) of the dielectric tensor using the response in the form (6.13) for magnetoelectric media; you may also need to use (5.5), (6.14) and (6.17).

6.6 Use the identity (2.23), *viz.*,

$$\begin{aligned}\epsilon_{abc}\epsilon_{ijk} = {} & \delta_{ai}\delta_{bj}\delta_{ck} + \delta_{ak}\delta_{bi}\delta_{cj} + \delta_{aj}\delta_{bk}\delta_{ci} \\ & - \delta_{bi}\delta_{aj}\delta_{ck} - \delta_{bk}\delta_{ai}\delta_{cj} - \delta_{bj}\delta_{ak}\delta_{ci},\end{aligned}$$

to derive an alternative form for (6.20) that does not involve a product of two permutation symbols.